Solution Design Document (SDD)

Project 5: Purchase Order Management System | Automation Anywhere A360

Purpose: Describe the proposed technical design for SAP PO status retrieval, real-time or scheduled status sharing, approved screenshot capture and timeout recovery. Exact SAP transactions, selectors and access mechanisms must be confirmed during discovery.

1. Solution Overview

An Automation Anywhere A360 bot retrieves PO status from the approved SAP source, validates and formats the records, optionally compares them with the previous successful snapshot, and distributes updates to approved stakeholders. Timeouts and unexpected application states are handled explicitly. Screenshot capture is limited to approved diagnostics or evidence.

2. Proposed Architecture

Component	Responsibility
A360 Control Room / scheduler	Schedules runs, manages deployment and provides execution monitoring.
Main orchestration bot	Loads configuration, coordinates the process and determines final run status.
SAP retrieval module	Handles authorized login, PO search/report execution and extraction.
Validation module	Checks required fields, normalizes values and applies status and duplicate rules.
Change comparison module	Optional comparison with the previous successful snapshot to identify changed statuses.
Screenshot capture module	Captures approved diagnostic evidence for defined exception conditions, subject to policy.
Report and notification module	Creates status reports and sends approved notifications.
Credential vault	Supplies credentials securely at runtime.
Logging and monitoring	Records run outcomes, retries, exceptions and delivery results.

3. Technical Workflow

1. Start the bot and create a unique run ID and timestamp.
2. Load and validate SAP connection settings, PO criteria, recipient list and output locations.
3. Open an authorized SAP session using secure credentials.
4. Execute the approved PO search or report and wait for a confirmed completion state.

5. Extract the required PO status fields.
6. Validate mandatory values, normalize formats and apply the approved duplicate key.
7. Compare against the last successful snapshot if change-only reporting is enabled.
8. For configured exceptions, capture minimal diagnostic evidence if permitted by security policy.
9. Create the status report and exception output.
10. Save outputs to the approved secure location.
11. Distribute only validated status information to the authorized recipients.
12. Record the outcome of each PO/search, delivery result and overall run status; close the SAP session.

4. Suggested Data Fields

Field	Purpose
Run_ID	Unique identifier for the automation run.
PO_Number	SAP purchase order number.
PO_Item	Item-level identifier where needed.
Supplier_ID / Supplier_Name	Supplier reference where authorized and required.
Company_Code	Company code associated with the PO.
PO_Status	Status retrieved from SAP or an approved status mapping.
PO_Date	Purchase order date, where available.
Last_Changed_At	Source update timestamp where available.
Retrieved_At	Timestamp of successful retrieval.
Change_Flag	New, changed, unchanged or not applicable when comparison is enabled.
Retrieval_Status	Success, zero results, timeout, failed or partial.
Exception_Detail	Sanitized error or reason for manual review.
Screenshot_Reference	Approved secure reference to diagnostic evidence, if captured.

5. Proposed A360 Bot Structure

Bot / task	Responsibility
BC_Main_PO_Status	Orchestration, configuration checks, record loop and final status.
BC_Login_SAP_ReadOnly	Secure login and session readiness validation.
BC_Retrieve_PO_Status	Runs the approved SAP search/report and extracts

	status data.
BC_Validate_PO_Data	Validates fields and applies mapping and duplicate rules.
BC_Compare_PO_Changes	Optional comparison against the previous successful snapshot.
BC_Capture_Exception_Evidence	Captures approved screenshots for defined error scenarios.
BC_Build_PO_Report	Creates the status report and exception list.
BC_Distribute_PO_Updates	Shares approved reports or notifications and records delivery.
BC_Log_And_Notify	Maintains run logs and sends support alerts.

These names are suggested conventions and can be adapted to the team's existing Automation Anywhere standards.

6. Timeout and Retry Design

- Use configurable timeouts for login, search, report execution and page transitions.
- Wait for a positive completion condition rather than relying only on a fixed delay.
- Retry only recoverable actions, with a finite retry count and appropriate wait interval.
- Do not repeat a transaction if doing so could create side effects; this workflow should remain read-only.
- If retries are exhausted, mark the relevant check as failed, preserve the error context and continue with other records where safe.
- A failed search must never be reported as a successful check with no updates.
- Record timeout counts and duration so support can identify recurring SAP performance issues.

7. Screenshot Capture Design

- Capture screenshots only for predefined conditions, such as an unexpected screen or an unresolved application error.
- Avoid capturing screens containing passwords, authentication codes or unrelated confidential information.
- Use the approved storage location and access permissions for screenshot files.
- Use a naming pattern such as RunID_Scenario_Timestamp, without including sensitive data.
- Store a reference to the screenshot in the exception log rather than embedding sensitive content in email notifications.
- Apply the approved retention period and deletion process.
- Disable screenshot capture where security or privacy rules prohibit it.

8. Real-Time Status Sharing

The exact meaning of 'real-time' must be confirmed with the process owner. If SAP does not provide an approved event-driven interface, the solution may use scheduled polling or on-demand retrieval. The reporting channel should clearly state the retrieval timestamp so recipients can judge how current the status is.

- Use event-driven updates only if an approved SAP interface supports them.
- For polling, configure an approved interval that respects SAP capacity and access policies.
- Send change-only notifications only after a successful comparison with the last successful snapshot.
- Avoid sending duplicate alerts for unchanged statuses unless requested by the business.
- Log notification delivery outcomes and provide a safe fallback route.

9. Security and Audit Controls

- Use a dedicated authorized account with read-only SAP permissions where possible.
- Store credentials in an approved vault or Control Room credential facility.
- Do not hard-code credentials or write passwords, tokens or authentication data to logs.
- Restrict reports and screenshot evidence to authorized users.
- Minimize supplier, purchasing and financial data in notification messages.
- Record run ID, timestamps, PO counts, retrieval outcomes, retries and distribution status.
- Apply approved encryption, retention, backup and secure deletion policies.

10. Logging and Monitoring

Event	Minimum details
Run start/end	Run ID, bot version, environment and timestamps.
SAP retrieval	PO selection reference, status, duration and record count.
Timeout/retry	Step, attempt number, elapsed time and sanitized error.
Validation	Valid, incomplete, duplicate and exception counts.
Screenshot capture	Trigger reason, secure file reference and capture status; avoid storing sensitive content in logs.
Report generation	Output location, record count and generation status.
Notification	Recipient group or destination and delivery result.

11. Testing Strategy

Test type	Coverage
Functional	Correct PO status extraction and field mapping for known test records.
Timeout	Slow response, stalled report, retry exhaustion and recovery.
Negative	Invalid credentials, access denied, missing PO and unexpected status.
Change tracking	New, changed and unchanged status behavior if

	enabled.
Screenshot policy	Correct trigger, safe content, secure storage and access restrictions.
End-to-end	Retrieval, validation, report creation, distribution and logging.
Security	Credential masking, read-only permissions and output access.
UAT	Procurement confirms the status report and exception process are fit for use.
Outcome measurement	Compare follow-up dependency against the agreed baseline to assess the 40% target.

12. Deployment and Support

13. Confirm SAP access method, PO scope, status fields and screenshot policy.
14. Build the bot and test components in the development environment.
15. Complete functional, timeout, security and end-to-end testing.
16. Obtain Procurement and IT/security approvals.
17. Promote the approved package and configuration through change management.
18. Monitor initial runs and verify reporting, exception alerts and delivery.
19. Hand over runbooks, retry settings, manual fallback and support ownership.

13. Design Items to Confirm

- SAP transaction/report or approved API/interface.
- Definition of real-time and required update frequency.
- Final PO fields, status mapping and duplicate key.
- Screenshot triggers, storage, permissions and retention.
- Timeout and retry values, alert recipients and fallback process.
- Report format, recipient groups and notification rules.
- Measurement method for the stated 40% reduction in follow-up dependency.

14. Definition of Done

- Approved SAP PO status retrieval works for in-scope records.
- Timeouts, failed checks and zero-result searches are distinguished correctly.
- Screenshot capture follows the approved security policy.
- Validated updates reach authorized recipients and delivery is logged.
- UAT, security approval, deployment sign-off and support handover are complete.